

TRAVELOS — CORPORATE TRAVEL MANAGEMENT SYSTEM

End-to-End System Flow, Technical Wireframes & Architecture Blueprint

STAKEHOLDERS

1

Traveling Employee

Raise · Track · Upload · Close

2

Approving Manager

Review · Remark · Decision

3

Travel Team (Desk)

Bookings · Timeline · Assets

4

Financial Team

Audit Claims · Credit Sal

5

Risk & Admin

Policy · SOS · Governance

SYSTEM FLOW (END-TO-END LIFECYCLE)

Phase 1: Landing & Login Screen

Auth Module

[Emp Dash Login Screen]

ID: EMP001 / EMP002

PW:

Log In → JWT Bearer

[Elevated Role Cards]

- **Manager:** Passcode 111111
 - **Travel Desk:** Passcode 222222
 - **Finance:** Passcode 333333
 - **Admin:** Passcode 444444
- * Instant role routing without ID input.

Phase 2: Employee Dashboard

Employee View

[Profile Summary]

- **Name:** Arjun Sharma
- **DOJ / Dept:** Active (Engg)
- **Govt ID:** Verified
- **Docs:** Attached

[Trip Registry]

+ Raise Travel Request

Active Trips	Closed Trips	Upcoming
TRIP-1024 (Live)	TRIP-0982 (Done)	TRIP-1055 (Pending)

Access live tracking, voucher downloads & closure triggers.

Phase 3: Create & Submit Request

Form Input

[Raise Travel Request Form]

- **Project & Client:** Project #B492 · Client #C-901
- **Dest & Duration:** Mumbai → London (7 Days)
- **Needs:** ☒ Flight ☒ Hotel ☒ Cab ☒ Extra Luggage
- **Asset Allocation:** ThinkPad X1 + Testing Device
- **Estimated Cost:** Total Budget: \$3,450

[Submit Request] → PENDING_AUTO_VAL

Phase 4: Automatic Validation

5-Check Engine

[AUTOMATIC VALIDATION]

- ✓ Is Emp Active?
- ✓ Does Project Exist?
- ✓ Is Valid Client?
- ✓ Budget Available?
- ✓ Mandatory Details & Company Policy OK?

✓ Yes → PENDING_MGR

✗ No → REJECTED_SYS

Phase 5: Manager Review

Manager

[Manager Dashboard Queue]

Pending: 3 | Approved: 14 | Rejected: 1

- **Audit Review:** Business purpose & budget fit
- **Asset Check:** Verify hardware allocation
- **Remarks:** Mandatory feedback input

[APPROVE]

[REJECT]

Status updates to APPROVED on sign-off.

Phase 6: Travel Desk & Booking

Travel Desk

[Travel Team (Travel Desk) Execution]

- **Flight Booking:** PNR: AI-204 · Dep: 08:00 AM · Arr: 11:00 AM
- **Cab Assignment:** Cab: DL-01-AB-1234 (Ramesh Kumar)
- **Hotel Reservation:** Grand Hyatt (Booking ID: HBK-90234)
- **Asset Handover:** HW-Asset-8492 logged & dispatched
- **Checklist Timeline:** Pre-fills & schedules all 7 milestone times

Activate Itinerary → Status: [ACTIVE]

Phase 7: Active Trip & Live Tracking

Live Verification

[7-Milestone Timeline with Mandatory Verification]

1	2	3	4	5	6	7
Flight Board	Flight Land	Cab Pickup	Hotel In	Hotel Out	Return Fit	Journey End
📄 Seat No	📄 Terminal	📄 OTP/Code	📄 Book ID	Pending	Pending	Pending

* Milestone accuracy: On-Time / Delayed / Early tracked vs Travel Desk schedule.

Phase 8: Financial Expense Settlement

Finance (Accounts)

[Financial Team (Accounts) Review]

- **Upload Bills:** Employee uploads receipts (Meals, Cabs, Emergency).
- **Audit & Verification:** Accounts team reviews claim categories & bills.
- **Salary Reimbursement:** Approved amount is CREDITED to employee salary.

* Approved expenses transition to status 'CREDITED' prior to trip closure.

Phase 9: Trip Closure & Reconciliation

4-Condition Gate

[Close Trip Automated Verification]

- ① **Trip Journey Ended:** Milestone 7 verified with arrival proof.
- ② **Assets Returned:** Physical hardware logged as returned.
- ③ **Expenses Settled:** All expense claims in CREDITED state.
- ④ **Dates & Policy:** Current date is past scheduled end date.

All 4 Verified → Finalize

[CLOSE TRIP] → Status: CLOSED

Frontend: Angular 16+ · Glassmorphism · RxJS Backend: Spring Boot 3.2 · Java 17+ · Spring Security Security: JWT Bearer · Role Guards Database: H2 File-based · Hibernate JPA Specs: 24 Endpoints · 8 Entities